OPEN MATERIAL LOGIC AND ADAPTIVE SYSTEMS LABORATORY

PART II — From Thresholds and Gates to Self-Regulating Flow Surfaces

A garage-scale experimental road from distinguishable relations to logic, memory, reconfiguration, feedback, and adaptive material organization.

BUILD • OBSERVE • MEASURE • MODEL • RECONFIGURE • REPEAT

No preferred ontology is required. Each stage earns only the claim it demonstrates.

OPEN REPLICATION INVITATION
This manual is designed so a capable beginner, maker, student, or independent laboratory can reproduce the relations directly. Successful, partial, contradictory, and failed replications are all useful. The aim is inspectable behavior before interpretive agreement.

0. Purpose, scope, and evidentiary posture

Part I demonstrated a low-barrier configured-material transduction experiment. Part II asks a different but continuous question: how far can simple, visible, reconfigurable relations be composed until the same substrate or coupled material assembly performs functions normally described as thresholding, logic, memory, routing, feedback, adaptation, and regulation?

The motivating research direction is explicit: progressively relocate sensing, state, discrimination, memory, routing, adaptation, and optimization from a separate external controller into the configured relations of the system itself. This is a research attractor, not an assumed conclusion.

> DISTINGUISH → RELATE → REPEAT → MEASURE → PREDICT → RECONFIGURE

LANGUAGE RULE
Terms such as material, information, logic, computation, memory, intelligence, controller, and environment are useful descriptions. This manual does not require their category boundaries as premises. The reproducible relation comes first; models and interpretations compete afterward.

Three levels of conclusion

Level	What may be claimed
Observed	A specified configuration under specified conditions produces a reproducible input-state-output relation.
Predictively modeled	A mathematical description forecasts measurements within stated error and operating limits.
Interpretive	Terms such as computation, memory, self-regulation, or optimization are proposed descriptions of the reproduced behavior.

The experimental ladder

> Threshold → Gate → Gate Network → Memory → Conditional Routing → Reconfiguration → Feedback → Adaptive Surface → Operating-Band Regulation

Every rung has four requirements: (1) a build that can be inspected, (2) minimal mathematics that predicts the relation, (3) a measurement protocol, and (4) a failure condition. A participant may stop at any rung without accepting the vocabulary used later.

1. Safety and laboratory limits

- Use only low-voltage electronics (recommended 3.3–12 V DC) for sensing and readout.
- Do not use high-speed turbine rotors. The culminating aerodynamic experiment is a stationary foil/blade section in low-speed airflow.
- Use guards around fans. Keep fingers, hair, cords, smoke sources, and loose materials away from fan blades.
- Use fog, theatrical haze, incense only where lawful and ventilated; water mist or tufts are preferred for beginner flow visualization.
- Magnets can pinch and can affect medical implants and electronics; use small craft magnets only.
- Spring steel, PET strips, and bistable elements can snap. Wear eye protection during adjustment.
- If chitosan or other chemical-responsive elements from Part I are incorporated, follow their separate chemical handling instructions.

SCOPE BOUNDARY

This manual does not instruct construction of an operating power turbine. The flow demonstrator uses a stationary or slowly articulated blade/foil in a guarded low-speed test stream.

2. Garage-scale primitive library

The system does not depend on one privileged material. Choose accessible substrates whose behavior can be seen, measured, and reset.

Primitive	Cheap realizations	State/variable	Use
Compliance	PET strip, spring steel, silicone, thin plywood, card	deflection q, stiffness k	gain, filtering, force redistribution
Threshold	over-center strip, magnet detent, buckled beam, elastic stop	critical load Fc	binary/conditional transition
Inversion	normally-closed contact, rocker, crossed linkage	sign/contact state	NOT/NAND/NOR
Memory	bistable strip, magnet latch, snap dome	state s	retention/hysteresis
Routing	string, lever, compliant beam, conductive tape	connectivity G	fan-out and conditional paths
Slow adaptation	chitosan/hydrogel, thermal strip, creep element	parameter p(t)	threshold retuning
Readout	LED + resistor, continuity meter, microswitch, Hall sensor	Y(t)	visible or logged output
Flow coupling	foil skin, rib, tab, flexible trailing edge	pressure, twist, camber	environment-system feedback

The minimal state description

$$y(t) = H[p(x), r(x,t), s(x,t), \text{boundary conditions}]$$

Here p describes configured parameters, r the imposed stimulus, s retained or evolving state, and y an observable output. This notation is descriptive: it does not declare one category more fundamental than another.

$$G = (V,E)$$

When connections matter, use a graph. V denotes distinguishable functional regions or states; E denotes permitted couplings or routes. Changing geometry may change parameters while preserving G; changing connectivity changes G itself.

3. Build the reusable two-input test frame

Before building individual gates, construct one reusable frame so the same parts can be rearranged. A 300 × 200 mm plywood, acrylic, or thick-card base is sufficient.

- Drill or punch a 20–25 mm grid of anchor holes, or use a pegboard/offcut with existing holes.
- Provide two input levers A and B with repeatable 0/1 positions.
- Provide a central replaceable threshold element.
- Provide a movable output contact or flag.

- Use screws, binder clips, magnets, or pegs so anchor position and prestress can be changed without rebuilding the frame.
- Mark a coordinate grid on the base. Record anchor coordinates for every configuration.

$$F_{eff} = w_A A + w_B B$$

$$Y = H(F_{eff} - F_c)$$

The weights wA and wB arise from lever arms, spring stiffness, geometry, and coupling. Fc is the transition threshold. The same physical cell can therefore implement different truth tables when these parameters are moved.

4. Gate 0 — YES / buffer

Build: one input lever deflects a compliant strip until it closes a visible contact or moves a flag across a reference line.

$$Y = H(A - A_c)$$

Test A=0 and A=1 for at least 20 cycles. Record missed transitions and the displacement margin around Ac.

5. Gate 1 — NOT / inversion

Use the same strip with a normally-closed output contact. Input displacement opens the route.

$$Y = 1 - H(A - A_c)$$

WHAT HAS BEEN DEMONSTRATED
A reproducible inversion relation. No claim about 'computation' is required yet.

6. Two-input AND and OR from one threshold cell

Couple A and B to the same threshold element. Keep the output normally open.

AND configuration

$$w_A < F_c,\ \ w_B < F_c,\ \ \text{but}\ \ w_A + w_B > F_c$$

$$Y = A \wedge B$$

OR configuration

$$w_A > F_c\ \ \text{and}\ \ w_B > F_c$$

$$Y = A \vee B$$

Reconfigure by moving anchor points, changing lever arms, changing spring preload, or changing the threshold element. Do not change the interpretation of the output after seeing the result: write the intended truth table first.

7. NAND and NOR

Retain the AND or OR mechanical configuration and reverse the readout topology with a normally-closed contact.

$$NAND = \neg(A \wedge B)$$

$$NOR = \neg(A \vee B)$$

This is an important separation: the threshold relation and the readout relation can be independently configured. One substrate need not contain a unique immutable gate identity.

8. XOR and XNOR

XOR requires distinguishing one active input from two. A garage-scale implementation can use a centered rocker or differential beam. Either A or B alone tips the rocker enough to close the output; equal simultaneous inputs drive the rocker toward the center or a second stop.

$$XOR: \; Y = A \oplus B$$

$$XNOR: \; Y = \neg(A \oplus B)$$

- Use equal lever arms initially.
- Tune the center restoring spring or magnet so 10 and 01 cross the output threshold.
- Tune the symmetric 11 condition so it does not.
- Reverse the output contact to obtain XNOR.
- Record the mechanical margin between the four input states; XOR is usually the most tolerance-sensitive elementary gate in this kit.

9. Gate characterization protocol

A	B	Expected Y	Observed Y	Peak disp.	Force/load	Pass?
0	0					
0	1					
1	0					
1	1					

Repeat the full truth table at least five times in randomized order. Twenty complete tables are preferred for publication.

noise margin ≈ distance from measured state cluster to switching threshold

Report false positives, false negatives, drift, hysteresis, cycle dependence, and any input-order dependence. A gate that works only when operated in one sequence is stateful and should not be reported as memoryless combinational logic.

10. Cascading, fan-out, restoration, and why isolated gates are not enough

A useful circuit requires the output of one cell to drive the input of another without ambiguity. Mechanical and material systems lose amplitude, accumulate friction, and shift thresholds.

$$G\_path \approx \prod g\_i \quad ; \quad \tau\_path \approx \Sigma\, \tau\_i$$

For a chain of cells, gain may multiply while delay accumulates. Noise, friction, compliance, and impedance mismatch require separate measurement.

- Begin with two gates only.

- Use a lever or spring stage as a mechanical impedance transformer if the first output cannot reliably switch the second.
- If stored energy is released by a bistable second stage, report that explicitly: the upstream gate triggers rather than supplies all output energy.
- Measure fan-out: how many downstream inputs can one output drive before its state becomes ambiguous?

11. Half-adder

$$S = A \oplus B$$

$$C = A \wedge B$$

Mount the XOR and AND cells side-by-side and feed both from the same two input levers. The two visible outputs are SUM and CARRY. This is the first recommended demonstration that clearly exceeds a single gate.

12. Full-adder and ripple carry

$$S = A \oplus B \oplus C\_in$$

$$C\_out = AB + C\_in(A \oplus B)$$

Build from two half-adders plus an OR stage. Keep the system slow and visible. A hand-operated cycle of several seconds is acceptable. Only after one full-adder is reliable should two be cascaded into a two-bit ripple-carry adder.

$$t\_total \approx N \cdot t\_stage$$

The delay scaling is itself an experimental lesson: computation has a geometry, a path length, and a propagation time.

13. Multiplexer and conditional routing

$$Y = (\neg S)A + SB$$

A selector S changes which route reaches the output. Mechanically this can be a rocker that couples either A or B to the output link. This experiment makes routing topology visible: the operation depends not only on state values but on which relations are enabled.

14. Bistability, hysteresis, and memory

Use an over-center spring strip, snap dome, magnetic detent, or buckled beam with two stable positions. Mark them 0 and 1.

$$U(q) \text{ has two local minima: } q0 \text{ and } q1$$

$$S(t^+) = F[S(t^-), I(t)]$$

Demonstrate that a brief input changes state and that the state persists after the input is removed. Measure retention time and the loads required to SET and RESET.

15. SR latch

A two-element mechanical latch can be made with crossed inhibitory links or two magnetically latched rockers. SET drives Q high; RESET drives Q low; the retained state supplies the next-cycle input.

$$Q_{n+1} = F(S, R, Q_n)$$

DO NOT OVERCLAIM
Persistent state is sufficient to demonstrate memory in the operational sense used here. It does not by itself establish learning, intelligence, or self-optimization.

16. Reconfigurable gate cell

Return to the original two-input frame and mark several anchor positions that produce distinct truth tables. The goal is one object whose operation changes by changing its configuration rather than replacing its material.

$$Y = F(X; p)$$

$$F_{p1} \neq F_{p2}$$

- Configuration A: AND.
- Configuration B: OR.
- Configuration C: NAND by output inversion.
- Configuration D: NOR.
- Optional differential configuration: XOR/XNOR.

Photograph and record the coordinates of every anchor, spring, contact, and stop. The configuration itself is part of the experimental specification.

17. Before Boolean logic: analog material operations

Boolean labels are imposed after thresholding. Many useful operations occur before the threshold and should be measured rather than discarded.

$$z = w_A A + w_B B$$

$$y = H(z - \theta)$$

The weighted sum z can itself be the output. Springs, beams, membranes, dampers, fluid restrictions, and thermal/hydration elements can implement weighting, filtering, integration, delay, rectification, and nonlinear saturation.

$$\tau \dot{x} + x = k\, u(t) \qquad \text{(first-order filtering / memory)}$$

$$m \ddot{q} + c \dot{q} + k q = F(t) \quad \text{(inertial-elastic response)}$$

This is the bridge from discrete gate demonstrations to continuous adaptive systems. The next experiments deliberately stop forcing every relation into 0 and 1.

18. Culminating demonstrator — the self-regulating flow surface

Build a stationary 300–500 mm foil or turbine-blade analogue in a guarded low-speed air stream. The objective is not a high-speed power turbine. It is a visible, measurable surface whose own configured compliance, thresholds, and retained states alter its aerodynamic interaction.

Suggested construction

- Rigid or semi-rigid leading-edge spar: wood dowel, aluminum strip, carbon tube, or printed spine.
- Three to five ribs: PET, thin plywood, printed polymer, or spring steel at selected hinge regions.
- Flexible skin: thin PET sheet, drafting film, fabric, or lightweight plastic.
- Trailing-edge torsion/compliance: elastic cord, spring strip, silicone hinge, or adjustable fishing-line tension.
- One bistable rib or magnet-detent element to create a retained high-load/low-load state.
- Optional slow-responsive insert: chitosan/hydrogel strip, bimetallic/thermal element, or another reversible material whose state shifts stiffness, curvature, or threshold.
- Visible tufts along the surface and wake; optional fog upstream for qualitative separation visualization.

Core coupled description

$$M(q)\,\ddot{q} + C(q)\,\dot{q} + K(q,s)\,q = F\_flow(U,p,q)$$

$$q \rightarrow \{\alpha\_eff,\ camber,\ contour\} \rightarrow \{p,\ \omega,\ separation\} \rightarrow q$$

q represents twist, camber, trailing-edge deflection, rib state, and other chosen coordinates. The surface changes the flow that loads the surface; the next state therefore depends on the previous coupled interaction.

19. First target behavior — passive load relief

$$U \uparrow \Rightarrow \text{aerodynamic load} \uparrow \Rightarrow \text{twist relief} \uparrow \Rightarrow \alpha\_eff \downarrow$$

$$\partial\alpha\_eff \,/\, \partial U < 0 \quad \text{over the designed operating range}$$

Tune the trailing-edge compliance so increasing airflow produces a modest twist or camber reduction. Compare with an otherwise similar rigid-control blade. The goal is not minimum deformation but bounded productive interaction.

20. Thresholded aerodynamic states

$$U < U1:\ C0 \text{ — efficient/nominal}$$

$$U1 \le U < U2:\ C1 \text{ — compliant relief}$$

$$U2 \le U < U3:\ C2 \text{ — stronger twist/depower}$$

$$U \ge U3:\ C3 \text{ — protective state}$$

These regimes can be implemented with sequential spring engagement, magnetic detents, over-center ribs, or staged stops. Measure the actual transition conditions rather than assuming the intended thresholds were achieved.

21. Productive middle region and operating-band regulation

The target is not zero interaction and not maximal interaction. A useful surface must retain enough form to structure the flow while remaining responsive enough for the flow to alter its state.

$$R_- < R^* < R_+$$

$$A = \{x : J(x) \geq J_{min}, \sigma < \sigma_{max}, \Gamma_{loss} < \Gamma_{max}\}$$

A is an experimentally defined viable operating region. J may combine useful force or torque proxy, drag/loss proxy, peak stress, and stability.

$$J = w1\ P_{useful} - w2\ D_{profile} - w3\ E_{sep} - w4\ \Gamma_{loss} - w5\ \sigma_{peak} - w6\ D_{fatigue}$$

The weights are engineering choices, not natural constants. Publish them. A different objective may select a different configuration.

22. Measuring the flow demonstrator

- Air speed U: handheld anemometer or calibrated fan setting.
- Useful force: inexpensive load cell or digital hanging scale in a repeatable fixture.
- Drag/load proxy: second load cell, spring scale, or calibrated deflection.
- Twist/camber q: printed angle scale, ruler, video tracking, or potentiometer.
- Surface state s: visual marker, microswitch, Hall sensor, or camera.
- Flow attachment/separation: tufts; optional fog or mist for qualitative visualization.
- Temperature/humidity if responsive materials are used.

$$\omega = \nabla \times u$$

Vorticity is not treated as inherently undesirable. Organized circulation can be essential to useful aerodynamic loading. The practical target is to reduce avoidable separated or dissipative structures while retaining useful momentum transfer.

23. Compare fixed and adaptive surfaces

Sweep airflow slowly through the same range for a rigid control and the adaptive surface.

$$J_{fixed}(U) \quad \text{versus} \quad J_{adaptive}(U)$$

$$\Delta U_{viable,adaptive} > \Delta U_{viable,fixed} \quad ?$$

A particularly strong result would be a wider useful operating range rather than a single higher peak. That directly tests whether the configured surface maintains productive interaction across changing conditions.

24. Environmental retuning

After the purely mechanical blade works, add one slow parameter that shifts a threshold or stiffness. Humidity-responsive chitosan is one accessible option, but it is not mandatory. Thermal strips, magnetic bias, fluid pressure, or other reversible elements are valid.

$$p = p(E,t)$$

$$F_c = F_c(p)$$

$$Y = F_{\{p(E,S)\}}(X)$$

Now identical airflow can produce different responses depending on retained or environmental state. Measure the shift in the transfer surface rather than describing it only qualitatively.

25. Closing the loop

$$E \rightarrow S \rightarrow C \rightarrow R \rightarrow C'$$

$$E_{t+1} = G(E_t,S_t)$$

$$S_{t+1} = F(S_t,E_t)$$

At this stage the distinction between system and environment becomes operational rather than absolute: the flow changes the surface, and the surface changes the subsequent flow.

26. Criteria for self-tuning and optimization

Do not call every responsive surface self-optimizing. First define an objective J and a configuration parameter p. Then demonstrate that the coupled dynamics alter p without an external controller calculating each successive setting.

$$p_{n+1} = p_n + \Delta p(E_n,R_n)$$

$$J(C_{n+1};E) > J(C_n;E) \quad \text{over a declared operating interval}$$

A stronger test changes the environment after adaptation. If the previous configuration becomes inferior and the system moves toward a different configuration, the preferred state is demonstrably conditional on environment.

$$C_A^*(E_A) \rightarrow C_B^*(E_B)$$

CLAIM DISCIPLINE
A single improvement event is not enough. Report repeated trajectories, failures, hysteresis, time constants, and whether the system can recover after the environment changes.

27. Noise, sensitivity, and common failure modes

Source	What it can mimic	Control
Friction/stiction	false thresholds, memory	repeat from both directions; lubricate or redesign pivots
Thermal drift	slow threshold shift	log temperature; allow equilibration
Humidity drift	stiffness/length changes	log RH; condition specimens
Shock/vibration	false switching	isolate table; repeat with dummy specimen
Magnet variability	threshold asymmetry	measure gap; use fixed mounts
Elastic fatigue/creep	apparent learning	cycle-control specimen; report cycle count
Contact bounce	multiple logic transitions	debounce mechanically or log raw contact
Fan turbulence	irregular switching	use flow straightener or average repeated sweeps
Fixture compliance	moving threshold	measure fixture motion; photograph setup

28. Replication and falsification standard

- Pre-write the intended truth table or response criterion before the run.
- Keep failed builds and failed cycles in the record.
- Randomize gate-input order where practical.
- Use matched fixed controls for the adaptive-surface experiments.
- Publish raw time series, not only selected peaks or successful videos.
- Report exact dimensions, anchor coordinates, materials, preload, environment, and cycle history.
- State what would count as failure before interpreting the result.

OPEN QUESTION TO REPLICATORS
If you reproduced the effect, what variables controlled it most strongly? If you could not reproduce it, where did the relation fail? If a simpler explanation accounts for the observation, demonstrate that explanation under the same conditions.

Appendix A — Bill of materials

Prices are approximate retail ranges and are intentionally non-brand-specific. Reuse household/shop items where available.

Item	Qty	Typical source	Approx. cost	Role
Base board / pegboard / acrylic	1	hardware/craft scrap	$0–15	reconfigurable frame
PET sheet / drafting film	several	office/craft	$5–15	compliant strips/skin
Spring steel strips or tape-measure stock	small pack	hardware/online	$5–15	threshold/bistability
Small compression/tension springs	assortment	hardware/online	$5–12	thresholds/preload
Small craft magnets	10–20	craft/online	$5–12	detents/latches
Binder clips, screws, washers, pegs	assorted	hardware/office	$5–15	anchors/fixtures
Fishing line / cord / elastic	1 pack	sport/craft	$3–8	routing/coupling
Copper tape / foil	1 roll	electronics/craft	$5–10	contacts/readout
LEDs + resistors + wire	small kit	electronics	$5–12	visible readout
Breadboard / low-voltage supply	1	electronics	$8–20	optional readout
Microswitches / Hall sensors	4–10	electronics	$5–15	state sensing
Digital scale or spring scale	1	retail/online	$10–25	force proxy
Load cells + amplifier boards	1–2	electronics/online	$10–30	quantitative force
Handheld anemometer	1	hardware/online	$15–30	air speed
Box fan / duct fan	1	household/hardware	$20–50	low-speed flow
Foil spar/rib materials	1 set	hardware/craft	$10–30	adaptive surface
Tuft yarn / light thread	1	craft	$2–5	flow visualization
Optional chitosan/hydrogel/thermal elements	small	lab/craft/online	$5–25	slow retuning

Practical target: about $60–120 for the gate laboratory if common tools and a fan are already available; about $120–250 for a more completely instrumented gate + adaptive-flow laboratory purchased largely from scratch.

Appendix B — Laboratory requirements and skill level

Requirement	Recommended level
Workspace	Stable table, good lighting, guarded fan area, basic ventilation.
Tools	Scissors, hobby knife, drill/awl, ruler/calipers, pliers, screwdrivers, clamps; soldering optional.
Electrical skill	Beginner. Only low-voltage continuity/LED/sensor circuits are required.
Mechanical skill	Beginner-to-intermediate: accurate cutting, repeatable fixtures, springs, levers, and careful adjustment.

Math skill	Algebra is enough to begin. Differential equations, state space, graphs, and optimization are introduced only when the experiment motivates them.
Data skill	Spreadsheet-level logging is sufficient. Video and CSV are strongly recommended for publication.
Safety	Eye protection for spring elements; fan guarding; no high-speed rotors or high voltage.

Appendix C — Configuration record sheet

Experiment ID: ____

Date/time: ____

Builder: ____

Gate / surface configuration: ____

Materials: ____

Anchor coordinates: ____

Prestress / spring settings: ____

Environmental conditions: ____

Input protocol: ____

Expected relation: ____

Observed relation: ____

Failures / anomalies: ____

Files / video / CSV names: ____

Appendix D — Suggested publication data schema

```
time, A, B, selector, state, output, displacement, force, U_air, T, RH, configuration_id, notes
```

Use explicit units in the header or metadata. Keep raw and processed files separate. Never overwrite failed runs.

Appendix E — Scientific context and provenance

This manual is aligned with the companion research program Field-Programmed Chitin Metamaterials: Research Program Edition 1.0 and the working paper Geometrically Governed Chitin Interfaces. Those works motivate the treatment of geometry, topology, prestress, boundary conditions, transduction pathways, environmental state, and falsification as parts of the reproducible configuration.

The garage-scale constructions in this manual are proposed experimental implementations. They should not be represented as established performance claims until independently built, measured, and replicated.

THE ROAD IS THE EXPERIMENT

See a relation. Control it. Configure it. Combine it. Let it retain state. Let it reconfigure. Close the loop. Measure whether it improves.

response → threshold → logic → state → memory → reconfiguration → feedback → adaptation → optimization

No agreement about ultimate meaning is required to participate.
Build the relation. Measure it. Publish what happened.